

Can a counterexample to direct finiteness be hyperbolic?

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Abstract

This note describes a proposed affirmative answer using OpenAI’s October construction of a torsion-free group algebra that is not directly finite. The idea is to retain its algebraic identities and extend its diagram argument to obtain a linear isoperimetric inequality. We explain the construction, give the proposed extension, and compare it with the marked-line and Leavitt-algebra approaches to nonsoficity. The conclusion remains provisional: the argument needs independent checking, and an apparently conflicting published assertion about idempotents in positive characteristic must be reconciled.

1 The question and the proposed answer

Can one find a torsion-free word-hyperbolic group G for which $\mathbb{F}_2[G]$ is not directly finite? A sufficient algebraic certificate is

$$ab = 1, \quad ac = 0, \quad c \neq 0. \quad (1)$$

Indeed, $ba = 1$ would give $c = b(ac) = 0$. Such a group is necessarily nonsofic, since group algebras of sofic groups over fields are directly finite [7]. Nonsoficity alone does not establish failure of direct finiteness.

The most promising route is the labelled-graph construction in *A Torsion-Free Group Algebra That Is Not Directly Finite* (4 October 2026) [1]. Its algebraic identities depend on local parity conditions, while its topology is controlled by excluding certain configurations of labelled paths. The proposed argument below uses a stronger application of the same path-exclusion estimate to control disk fillings. If valid, a suitable choice of the original graphs would already give a hyperbolic group.

The distinction between the source and this extension matters. The October manuscript asserts finite presentability and torsion-freeness; hyperbolicity is the additional conclusion proposed here. The disk argument and its use of the source estimates are presented for review. There is also a specific literature issue: a positive-characteristic idempotent assertion of Bartels–Lück–Reich would exclude the proposed conclusion. Section 7 explains why this remains unresolved. This note was prepared with language-model assistance and has not received independent expert verification.

2 Why the algebra survives the choice of graphs

Let F be a rose whose oriented edges are labelled by a finite alphabet T , with inverse label $\bar{t}$. Let $\Gamma = \Gamma_A \sqcup \Gamma_B \rightarrow F$ be an immersion of finite loopless labelled graphs. Form X by attaching a cone on each connected component along its label map, and put $G = \pi_1 X$. Choose roots x_A, x_B , with components A', B' .

Write S_x for the set of outgoing labels at x . Immersion makes the endpoint $x \cdot t$ unique for $t \in S_x$. Labels of root-to-vertex paths define elements $g_x, h_y \in G$: these are independent of the path because all component loops are killed by the cones. In particular,

$$g_{x_A} = h_{x_B} = 1, \quad g_{x \cdot t} = g_x t, \quad h_{y \cdot t} = h_y t.$$

Proposition 1 (The parity identities). *Suppose $|S_x \cap S_{x'}|$ is odd for all $(x, x') \in A' \times A'$ except (x_A, x_A) , where it is even. Suppose also that $|S_x \cap S_y|$ is odd for every $(x, y) \in A' \times B'$, and that $h_y \neq 1$ for $y \neq x_B$. Then*

$$a = \sum_{x \in A'} g_x, \quad b = \sum_{x \in A'} g_x^{-1}, \quad c = \sum_{y \in B'} h_y^{-1}$$

satisfy $ab = 1$, $ac = 0$, and $c \neq 0$ in $\mathbb{F}_2[G]$.

Proof. On $A' \times A'$, join (x, x') to $(x \cdot t, x' \cdot t)$ for each common outgoing label. The degree is $|S_x \cap S_{x'}|$, and the contribution to ab is constant on every connected component, since

$$(g_x t)(g_{x'} t)^{-1} = g_x g_{x'}^{-1}.$$

By the handshaking lemma, a component with only odd-degree vertices has even order, so its contribution vanishes in characteristic two. The component containing (x_A, x_A) has exactly one even-degree vertex and hence odd order. It contributes 1. Thus $ab = 1$. The same argument on $A' \times B'$, where every degree is odd, gives $ac = 0$. Finally, the coefficient of 1 in c is one, by the assumption on the root. $\square$

The source arranges these parities using lines in the projective plane over $\mathbb{F}_{128}$, together with seven additional labels from a Fano-plane design. Ordinary line intersections are odd; the extra labels preserve that parity except at the distinguished root. Non-root degrees are 129 or 133, and the distinguished degree is 136. Each label has the same number of slots as its inverse. Pairing those slots defines the edges. Crucially, changing the pairings changes the geometry without changing the parity calculation. The separate topological argument is needed to keep the root term of c from cancelling; distinctness of all path labels is unnecessary.

3 The proposed linear filling bound

Proof idea. If a minimal disk filling had interior boundary much larger than its exterior word, then cutting the inner boundaries into bounded pieces and applying a planar separator would produce a bounded collection of paths that is almost completely paired. This is precisely the configuration excluded by the October manuscript's path estimate. Hence interior boundary length is linear in the exterior length, and therefore so is filling area. The constants below make this extraction precise.

All lengths in this section count edge traversals. A comparison pairs two disjoint intervals of positions carrying inverse words in reverse order; each paired pair of positions must traverse *distinct underlying unoriented graph edges*. Repeated traversals elsewhere are allowed. Comparisons use disjoint positions.

Assume every component of Γ has girth at least $L \geq 3$, diameter at most dL , and minimum degree at least 129. Assume there is a fixed short-closure constant D_0 : every immersed segment can be extended, with no original cancellation, to a cyclically immersed closed path by appending at most $D_0 L$ edges.

For fixed $\varepsilon \in (0, 1/4]$, put

$$\begin{aligned} C_{\text{sep}} &= \frac{2\sqrt{2}}{1 - \sqrt{2/3}}, & U &\geq \max\{3, \lceil 96D_0/\varepsilon \rceil\}, \\ \eta &= \min\{1/(32U), \varepsilon/(64U)\}, & K_0 &= \lceil (C_{\text{sep}}/\eta)^2 \rceil, \\ C &= K_0(U + D_0), & I &= \lceil 192C \rceil. \end{aligned}$$

Suppose there is no system of at most K_0 positive-length, cyclically immersed closed paths, with total length H_Z satisfying

$$L \leq H_Z \leq CL,$$

at most I comparisons as above, and at most εH_Z unpaired positions. Call this the *closed bounded-system exclusion*. This is a finite, decidable property of a finite graph.

Theorem 2 (Disk estimate). *Under these finite hypotheses, the cone group G is word-hyperbolic. Choose one maximal tree in each component and take its fundamental cycle labels as relators on the rose generators. In this finite presentation, every nonempty cyclically freely reduced nullword obeys*

$$\text{Area}(w) < \frac{64}{\varepsilon} |w|.$$

Proof. Step 1: choose a minimal filling. Let w be a nonempty cyclically freely reduced nullword, $n = |w|$. Replace cone fillings by the fundamental cycle relators, or use a standard picture representation. A disk filling can be represented by finitely many inner disks, each with a closed boundary lift in a component of Γ , and a complementary surface mapping to F . The outer boundary remains the exact rose word w ; it is not assumed to lift to Γ . Minimize the sum H of the inner boundary lengths. Tighten each inner path cyclically and discard constant inner disks. At least one positive inner disk remains, since w cannot bound in the rose. Thus $H \geq L$.

Step 2: remove cancellable inner pairings. General position over one interior point of every rose edge pairs all boundary occurrences by disjoint embedded arcs; closed preimage curves may be discarded. Ends of each arc have inverse labels. If an arc joins occurrences of the same underlying graph edge on two inner disks, merge them through their common cone and delete the two edge occurrences. If it joins one inner disk to itself, its disk and band form an annulus. Keep the complementary component containing the outer boundary, and cap its new inner boundary in the same abstract cone. The exterior filling still exists, and inner length decreases by at least two. Tightening can only decrease it further. These surgeries contradict minimality. Therefore all inner-inner pairs use distinct underlying graph edges. These surgeries leave the outer word unchanged and require lifts only of inner boundaries.

Step 3: divide long paths into bounded pieces. Suppose for contradiction $n \leq \varepsilon H/64$. Keep each inner loop of length at most UL intact. Split each longer one into immersed segments with lengths between $UL/3$ and UL ; append short closures. Let N be the number of resulting pieces. Every original piece has length at least L , so $NL \leq H$. The total appended length A is bounded by

$$A \leq \frac{3D_0H}{U} \leq \frac{\varepsilon H}{32}.$$

Only original occurrences retain pairings; all closure positions are unpaired.

Step 4: count comparisons, including the exterior. Regard the outer boundary as one additional formal disk. The planar paired-interval count used in the source [1] gives at most $6(N+1)$ interval pairs. Its proof only uses inverse-letter pairings and cyclic free reduction to exclude monogons and self-paired runs; it does not use graph lifts. The count comes from thickening disks and pairing arcs into bands: digon regions continue a comparison, while Euler characteristic bounds the remaining gaps. Cutting at these gaps, at indexing points, and at the segment boundaries leaves at most $12(N+1)$ runs, paired in distinct twos. Thus

$$\#\{\text{comparison intervals}\} \leq 6(N+1) \leq 12H/L.$$

Delete the exterior vertex and its incident arcs. At most n inner positions lose partners.

Step 5: extract a small forbidden collection. The pairing graph on the N inner pieces is planar: pieces are consecutive boundary intervals and can be separated within their old disks without crossing external bands. The recursive planar separator bound [8] removes at most

$C_{\text{sep}}N/\sqrt{K_0} \leq \eta N$ piece vertices and leaves components with at most K_0 vertices. The removed *original* length M satisfies

$$M \leq \eta NUL \leq \eta UH \leq \varepsilon H/64.$$

Surviving pieces lose at most M additional partners.

For a surviving component Z , include its closures in H_Z , write b_Z for unpaired positions, and I_Z for comparisons. Summing over components gives

$$\begin{aligned} \sum_Z H_Z &\geq H - M \geq 63H/64, \\ H_Z &\leq K_0(U + D_0)L = CL, \\ \sum_Z b_Z &\leq n + M + A \leq \varepsilon H/16, \\ \sum_Z I_Z &\leq 12H/L. \end{aligned}$$

Components with $b_Z > \varepsilon H_Z$ have total length at most $H/16$. Components with $I_Z > 192H_Z/L$ have total length at most $H/16$. Some positive length survives both deletions, indeed at least $55H/64$. Such a component consists of positive cyclically immersed loops, so its length is at least L by girth. It is an excluded system: $H_Z \leq CL$, $b_Z \leq \varepsilon H_Z$, and $I_Z \leq 192C \leq I$. Contradiction. Hence $H < 64n/\varepsilon$.

Step 6: convert boundary length to filling area. For each component choose a tree root, tree paths t_v , and relators $t_{\text{init}(e)}e t_{\text{term}(e)}^{-1}$ for non-tree edges e . A closed component path of length h , after conjugating to the tree root, is a product of at most h such relators or their inverses: insert the tree paths at successive vertices and telescope. Tree-edge factors freely cancel. Thus each inner disk costs at most its boundary length, and the complementary rose surface adds no relation cells. Total area is at most H . Free or cyclic reductions and whiskers add no relation cells, so the same linear inequality applies to arbitrary nullwords after reduction. A finitely presented group with a linear Dehn function is word-hyperbolic. $\square$

Two features of the proposed extension deserve particular scrutiny. The outer word need not lift to Γ , so same-edge surgery is used only on inner boundaries. Also, excluding fully paired spheres alone is insufficient: the argument needs the stronger exclusion of *nearly* fully paired bounded collections.

4 Selecting the same graphs for algebra and geometry

The order of quantifiers in the source estimate is essential. The October manuscript [1] first chooses $\varepsilon > 0$. For every subsequently fixed triple (K_0, C, I) , it asserts that the prohibited path configurations occur with probability $o(1)$, under its matching law conditioned on girth at least $L = \lfloor c_0 \log n \rfloor$, as the graph size n grows. The configurations include the closed collections above and also allow one immersed interval beginning at either designated root, x_A or x_B .

The source's switching argument also asserts that the girth-conditioned law is nonempty. The same law is asserted to give component diameter $O(L)$ with probability $1 - o(1)$. Together with the minimum-degree bound, this supplies the short-closure constant D_0 . We may therefore fix all the constants in Theorem 2 before letting n grow. The construction does not require an estimate uniform over growing values of K_0, C, I .

There are two substantial probabilistic inputs here. First, a switching argument controls edge prescriptions *after conditioning on high girth*; ordinary independence cannot be assumed. Second, a weighted word count treats repeated traversals by multiplicity and gives exponential decay for every fixed bounded pattern. These are inputs from the source, rather than consequences of the disk estimate.

The source also uses path exclusion to rule out reduced spherical cone diagrams and to protect the root-to-vertex labels. Its topology argument gives $\pi_2(X) = 0$ and $h_y \neq 1$ for $y \neq x_B$. Because X is two-dimensional, its simply connected universal cover is then contractible. Thus X is a finite-dimensional classifying complex, which excludes torsion in G .

Corollary 3 (Conditional conclusion). *Assuming the stated source estimates and the disk argument above, one can choose the October labelled graphs so that G is torsion-free and word-hyperbolic, while $\mathbb{F}_2[G]$ is not directly finite.*

Proof. Fix the constants for the disk exclusion and those for the source’s rooted spherical exclusion. The diameter condition and both exclusions hold simultaneously with probability $1 - o(1)$ under the same conditioned matching law. Select one sufficiently large outcome. The topology argument gives torsion-freeness and protects c ; Proposition 1 gives (1); and Theorem 2 gives hyperbolicity. $\square$

This selects a stronger event than the original final assembly. It does not say that every graph satisfying the original spherical exclusion is hyperbolic. The construction remains existential: no specific graph of the required size, or expanded list of the elements a, b, c , is supplied here.

5 How the constructions differ

Labelled graphs and parity. In the October manuscript, parity cancellation is used to obtain the identities (1) in a group algebra. It develops the preceding zero-divisor construction [2]: both use cancellation in pairs of synchronously labelled paths. A special root changes one zero product into a product equal to 1. The probabilistic and topological arguments are used to show that the other factor c survives. This is stronger information than nonsocficity alone.

Marked lines and rank. The marked-line manuscript [3] uses punctured affine lines, Cartesian blocks of triangle relations, and a rank comparison. Its triangle complex has negatively curved links, giving geometric control independently of the rank argument. The original claim is a torsion-free hyperbolic group that is not residually finite. A separate, proposed extension to approximate permutation representations would establish nonsocficity: rectangle words force a rank lower bound, while incidence identities force an incompatible upper bound. That extension does not provide a, b, c satisfying (1).

Leavitt algebras and expansion. The Leavitt unit-group construction [4] uses a property-(T) subgroup, a commuting subgroup that cannot be locally approximated by finite groups, and elements that transport these subgroups into one another. Expansion obstructs sofic approximations. Although the ambient Leavitt algebra itself is not directly finite, a map onto that algebra does not lift a one-sided inverse to the group algebra of its unit group. Nor does this construction itself give a hyperbolic group.

The earlier September direct-finiteness constructions use another route: finite-group idempotents and matrix compression, followed by an embedding of a matrix algebra into a group algebra [5, 6]. These share rank and interpolation ideas with the marked-line construction, but their displayed presentations contain torsion. Thus there are shared ideas, especially finite-dimensional rank obstructions, without a known transformation that preserves the direct-finiteness certificate between the groups.

6 Why the other geometric routes are less direct

The marked-line complex obtains curvature from its vertex links. Applying that method to the original cone complex encounters a concrete obstruction. Repeated label types create many four-cycles in the rose link. The necessary angle inequalities force the total angle available at cone apices to be at most 68π . On the other hand, minimum graph degree 129 supplies at least 64 edge-disjoint cycles, each requiring apex-link length at least 2π . Consequently the original triangulation cannot be made locally CAT(0) merely by choosing Euclidean or hyperbolic shapes for its triangles. This concerns that triangulation; it does not rule out another geometric model for the group.

There is a similar limitation on a direct graphical small-cancellation argument. The fixed alphabet and high degree force long pairs of paths with the same label. A counting argument gives a limiting lower bound greater than 0.49 for the longest-piece/global-girth ratio, whereas Ollivier's global metric criterion requires it to be less than $1/6$ [9]. This does not rule out every cycle-relative graphical condition. It explains why the proposed disk argument uses a more specific exclusion of almost fully paired closed collections instead of a bound on all open-path overlaps.

Algebraic transfers require care as well. A homomorphism $G \rightarrow H$ preserves the two equalities in (1), but gives a counterexample in $\mathbb{F}_2[H]$ only if the image of c remains nonzero. Injectivity on its support is sufficient. Conversely, a hyperbolic group surjecting onto G does not lift the identities to its own group algebra. Neither a quotient argument nor a hyperbolic cover therefore settles the question without additional control.

7 The unresolved idempotent issue

The published hyperbolic-group paper of Bartels–Lück–Reich states a positive-characteristic trivial-idempotent consequence that would apply to $\mathbb{F}_2[G]$ for torsion-free hyperbolic G [10, p. 33]. Taken as stated, this excludes the proposed conclusion, because (1) gives the nontrivial idempotent

$$e = 1 - ba, \quad e^2 = e.$$

Indeed $ba \neq 1$, while $ba \neq 0$ follows from $a(ba)b = (ab)^2 = 1$.

There is a difficulty in following the cited justification. The general idempotent theorem in the companion paper requires an additional stable-finiteness, characteristic-zero, or soficity hypothesis [11, Theorem 1.12]. The relevant positive-characteristic statements in Formanek and Burger–Valette control traces; their full idempotent conclusions use characteristic-zero arguments [12, 13].

Trace information alone cannot exclude this particular idempotent. Since $ab = 1$, one has $e = ab - ba$. For every conjugacy class $\mathcal{C}$, the sum of the coefficients on that class satisfies

$$\mathrm{tr}_{\mathcal{C}}(ab) = \mathrm{tr}_{\mathcal{C}}(ba), \quad \mathrm{tr}_{\mathcal{C}}(e) = 0,$$

because gh and hg are conjugate. Likewise, for $R = \mathbb{F}_2[G]$,

$$R = baR \oplus eR, \quad baR \cong R, \quad [eR] = 0 \text{ in } K_0(R).$$

Thus these trace and K_0 conclusions already hold for any hypothetical counterexample; an additional argument is needed to deduce $e = 0$.

This does not establish that the published assertion is false. It identifies an unresolved conflict that must be settled before the affirmative conclusion can be accepted. The two central mathematical checks are therefore the fixed-exterior disk argument above and the precise scope of the positive-characteristic idempotent result, together with the source probability estimates on which the construction rests.

References

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